FURTHER MATHEMATICS UNIT 3
APPLICATION TASK SAC 1 2020

CORE – DATA ANALYSIS : PART 2

NAME: ________________________________ TEACHER: _______________

Dates: **PART TWO – Wednesday 22th July 3:30 – 5: 40 pm**

Duration: **PART TWO – 10 minutes reading time plus 120 minutes**

Materials allowed: **1 x Bound reference, CAS calculator, scientific calculator, ruler,**

pens/pencils Outline of SAC

At the end of each session the student must:

1. **Submit their SAC,**
2. **Sign to state that they have submitted all these items and that the work is their own.**

Total : 67 marks

PART ONE : 26 marks

PART TWO : 41 marks

STUDENT DECLARATION

- **I declare that the work completed in this assessment is my own work and demonstrates my own abilities and knowledge and does not involve plagiarism or collaboration with other students**
- **I declare that I have not knowingly assisted another student in a breach of rules**
- **I understand that I must not cheat or assist other students to cheat, including taking any action that gives or attempts to give me or another student an unfair advantage**
- **I declare that I have not engaged in unacceptable forms of assistance including; use of, or copying, another person's work or other resources without acknowledgement; corrections or improvements made or dictated by another person**
- **I declare that I have not contracted another person to do the work for me or allowed another person to edit and substantially change my work.**
- **So that the teacher can properly assess my work, I give my teacher permission to reproduce this work and provide a copy to another member of staff for the purpose of cross checking and moderation and to take steps to authenticate its originality.**
- **I understand that if a teacher is in doubt about the authenticity of my work, an authentication panel will conduct an investigation into any breach of VCAA rules**
- **I understand that any School-assessed Coursework marks given to me are conditional and may be altered if: the work is atypical of other work produced by me; is inconsistent with the teacher's knowledge of my ability; the work contains unacknowledged material; the work as not been sighted and monitored by the teacher during it development (if relevant)**
- **I understand that total scores for School-assessed Coursework may change as a result of statistical moderation**

Signed: ___________________________________

Date: _____________________________________

<u>In order to satisfactorily complete this SAC</u> students must demonstrate an understanding of the following three outcomes. These outcomes are covered throughout the SAC.
The following table summarises what each outcome entails in terms of key knowledge and skills. Please note that not all points are covered in this SAC- indicated by an NA (not assessed).

Outcome 1 (10 marks): On completion of this unit the student should be able to define and explain key concepts and apply related mathematical techniques and models as specified in Area of Study 1 in routine contexts. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1.	
Key knowledge	**Satisfactory**
❑ **types of data: categorical (nominal and ordinal) and numerical (discrete and continuous)**	
❑ **frequency tables, bar charts including segmented barcharts, histograms, stem plots, dot plots, and their application in the context of displaying and describing distributions**	
❑ **log (base 10) scales, and their purpose and application**	
❑ **five-number summary and boxplots (including the designation and display of possible outliers)**	
❑ **mean and standard deviation**	
❑ **normal model and the 68–95–99.7% rule, and standardised values (z-scores)**	
❑ **response and explanatory variables**	
❑ **two-way frequency tables, segmented bar charts, back-to-back stem plots, parallel boxplots, and scatterplots, and their application in the context of identifying and describing associations**	
❑ **correlation coefficient, r, its interpretation, the issue of correlation and cause and effect**	
❑ **least squares line and its use in modelling linear associations**	
❑ **data transformation and its purpose**	
❑ **time series data and its analysis**	
Key skills	
❑ **construct frequency tables and bar charts and use them to describe and interpret the distributions of categorical variables**	

❑ answer statistical questions that require a knowledge of the distribution/s of one or more categorical variables	
❑ construct stem and dot plots, boxplots, histograms and appropriate summary statistics and use them to describe and interpret the distributions of numerical variables	
❑ answer statistical questions that require a knowledge of the distribution/s of one or more numerical variables	
❑ solve problems using the z-scores and the 68–95–99.7% rule	
❑ construct two-way tables and use them to identify and describe associations between two categorical variables	NA
❑ construct parallel boxplots and use them to identify and describe associations between a numerical variable and a categorical variable	
❑ construct scatterplots and use them to identify and describe associations between two numerical variables	
❑ calculate the correlation coefficient, r, and interpret it in the context of the data	
❑ answer statistical questions that require a knowledge of the associations between pairs of variables	
❑ determine the equation of the least squares line giving the coefficients correct to a required number of decimal places or significant figures as specified	
❑ distinguish between correlation and causation	
❑ use the least squares line of best fit to model and analyse the linear association between two numerical variables and interpret the model in the context of the association being modelled	
❑ calculate the coefficient of determination, r^2, and interpret in the context of the association being modelled and use the model to make predictions, being aware of the problem of extrapolation	
❑ construct a residual analysis to test the assumption of linearity and, in the case of clear non-linearity, transform the data to achieve linearity and repeat the modelling process using the transformed data	
❑ identify key qualitative features of a time series plot including trend (using smoothing if necessary), seasonality, irregular fluctuations and outliers, and interpret these in the context of the data	
❑ calculate, interpret and apply seasonal indices	NA
❑ model linear trends using the least squares line of best fit, interpret the model in the context of the trend being modelled, use the model to make forecasts being aware of the limitations of extending forecasts too far into the future	

Outcome 2 (20 marks): On completion of this unit the student should be able to select and apply the mathematical concepts, models and techniques as specified in Area of Study 1 in a range of contexts of increasing complexity. To achieve this outcome the student will draw on knowledge and skills outlined in Area of Study 1	
Key knowledge	
❑ **the facts, concepts and techniques associated with data analysis**	
❑ **standard models studied in data analysis and their area of application**	
❑ **general formulation of the concepts, techniques and models studied in data analysis**	
❑ **assumptions and conditions underlying the use of the concepts, techniques, and models associated with data analysis**	
Key skills	
❑ **identify, recall and select facts, concepts, models and techniques needed to investigate and analyse statistical features of a data set with several variables that can include time series data**	
❑ **interpret and report the results of a statistical investigation or of completing a modelling or problemsolving task in terms of the context under consideration, including discussing the assumptions in application of these models**	
Outcome 3 (10 marks): On completion of this unit the student should be able to select and appropriately use numerical, graphical, symbolic and statistical functionalities of technology to develop mathematical ideas, produce results and carry out analysis in situations requiring problem-solving, modelling or investigative techniques or approaches. To achieve this outcome the student will draw on knowledge and related skills outlined in Area of Study 1.	
Key knowledge	
❑ **• the difference between exact numerical and approximate numerical answers when using technology to perform** ❑ **• computation, and rounding to a given number of decimal places or significant figures**	
❑ **domain and range requirements for specification of graphs of models and relations, when using technology**	
❑ **• the relation between numerical, graphical and symbolic forms of information about models and equations and** ❑ **• the corresponding features of those functions and equations**	
❑ **similarities and differences between formal mathematical expressions and their representation by technology**	

❑ the selection of an appropriate functionality of technology in a variety of mathematical contexts	
Key skills	
❑ distinguish between exact and approximate presentations of mathematical results produced by technology, and interpret these results to a specified degree of accuracy in terms of a given number of decimal places or significant figures	
❑ use technology to carry out numerical, graphical and symbolic computation as applicable	
❑ produce tables of values, families of graphs and collections of other results using technology, which support general analysis in problem-solving, investigative and modelling contexts	
❑ identify the relation between numerical, graphical and symbolic forms of information about models and equations and the corresponding features of those models and equations	
❑ select an appropriate functionality of technology in a variety of mathematical contexts, related to data analysis, and provide a rationale for these selections	
❑ relate the results from a particular technology application to the nature of a particular mathematical task (investigative, problem solving or modelling) and verify these results	
❑ specify the process used to develop a solution to a problem using technology, and communicate the key stages of mathematical reasoning (formulation, solution, interpretation) used in this process	

Overall result:	Outcome 1	Outcome 2	Outcome 3

After SACs have been marked, teachers will decide if a student has demonstrated enough knowledge throughout their SAC to be given a satisfactory. If a student can't demonstrate through her SAC that she understands the content covered then the student must demonstrate her knowledge by either: 1. Showing her teacher her up-todate workbook; and/or, 2. Completing a short test.

Part 2: 41 Marks

Introduction

When using the ATAR Calculator to decide which University courses they wish to apply for, students need to predict their study scores in different subjects. In order to assist students to do this a team of Mathematicians at Killester College are analysing the relationship between a student's SAC results and their Study Score.

The SAC results for a random group of 18 students from Killester 2019 Further Mathematics classes are used to complete this investigation.

In this task we will investigate the following:

- **Analyse a Year 12 Further Maths Class' students' SAC results.**
- **Query whether a relationship exists between the SAC results and the Study Scores?**
- **Conclude whether an accurate prediction of a student's Study Score in Further Mathematics can be made from their SAC results.**

Note: For the purposes of this investigation, the SAC results refer to the mean of the students' Further Maths SACs for the year, and are out of 100.

The initial task of the investigation

Before investigating the relationship between the SAC results and the Study Scores, the team of Mathematicians at Killester decide that they want to revisit the previous analysis of a FM class' SAC results in 2018 to have a better understanding of how this group of Killester students performed in their SACs.

The team found that the SAC results for this class were normally distributed with a mean of 66 and a standard deviation of 11 (SAC results were out of 100 and there were 25 students in this class).

1. Draw the bell-shaped curve representing the normal distribution for the SAC results for this class. Include values three standard deviations from the mean. (2 marks)

2. There were 25 students in this class. How many students had achieved a result greater than 77? (1 mark)

3. Two students are selected with the Z-score of 2.11 and 2.43.

(a) Are these two students in the top 2.5% of the group. Give reasons why? (1 mark)

(b) Calculate their actual SAC results. (2 marks)

(c) The SAC results for all students studying Further Mathematics in Victoria in 2018 are normally distributed with a mean percentage of 57 and a standard deviation of 14. In this 2018 group of Year 12 FM students at Killester, there were 5 students achieving a SAC result greater than 85.

Approximately what percentage of the class of 25 would be in the top 2.5% of the state? (2 marks)

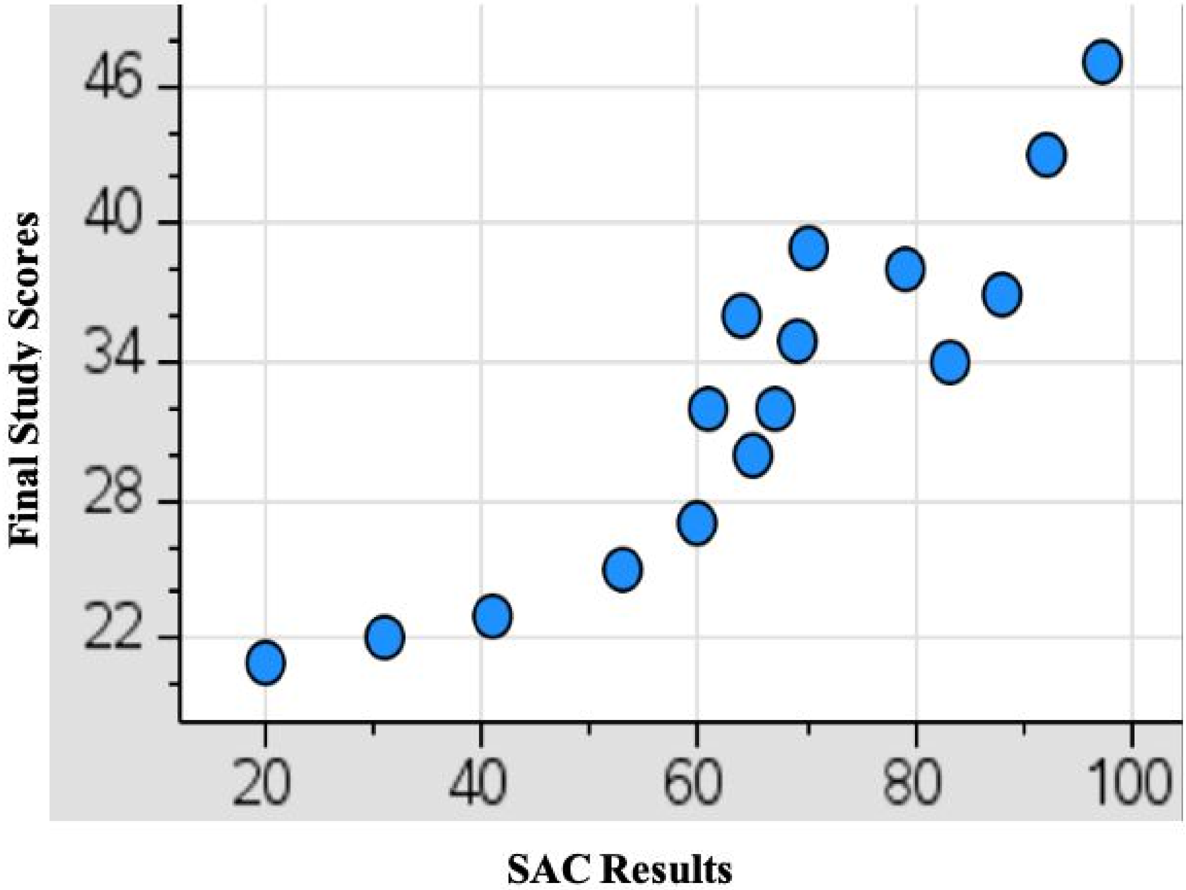
The next phase of the investigation

The team of Mathematicians at Killester has randomly selected a group of 18 students' SAC results and their Study Score from 2019 Further Mathematics classes, shown below.

Student Results:

STUDENT	SAC RESULT Score out of 100	STUDY SCORE Score out of 50
A	20	21
B	27	20
C	31	22
D	53	25
E	41	23
F	61	32
G	67	32
H	60	27
I	70	39
J	64	36
K	65	30
L	69	35
M	79	38
N	76	33
O	83	34
P	88	37
Q	92	43
R	97	47

4. A scatterplot of the data is provided but it is missing two pieces of data – student B and student N. Fill in their results on the scatterplot and label them clearly. **(2 marks)**



5. What is the form of this scatterplot? **(1 mark)**

6. What would be the problem with fitting a linear regression model to this data? **(1 mark)**

7. Which transformations should be considered in order to linearise this data? **(1 mark)**

8. Complete the following table using the transformations you chose in question 7. (3 marks)

Transformation	Coefficient of determination (to 3 significant figures)	Equations in terms of SAC results and Study Score (to 3 significant figures)

9. Which transformation of data would provide the highest predicted Study Score for students? Explain your chosen model using student L's SAC result of 69. (2 marks)

The team of Mathematicians at Killester College has decided to choose the model below.

$$\textbf{Study score} = \mathbf{19.6 + 0.00274 \times \textbf{SACs Result}^2}$$

The model above will apply to the following **questions 10 - 14.**

10. Describe the association between the SAC results and the Study Scores in terms of strength and direction. (1 mark)

11. What percentage of the variation in the Study Score can NOT be explained by the variation in the SAC result. (Give your answer correct to 1 decimal place) (1 mark)

12. A student who achieves 62 on SACs wants to use this model to predict her Study Score.

(a) Find her predicted Study Score, correct the nearest whole number. (1 mark)

(b) Comment on the reliability of this prediction. (1 mark)

13. Allysa was aiming for a Study Score of 40. Using the provided model, what mean mark must she achieve on her SACs to the nearest whole number? (1 mark)

14. a) What is the residual for Student J to 2 decimal places? (1 mark)

b) What does this residual mean? (1 mark)

Establishing a Maths Enhancement Centre

To help support Further Mathematics students to do well in their SACs and the end of year examinations, a drop-in centre called Maths Enhancement Centre has been established. They provide one-on-one help with students out of school hours as well as provide the option to purchase summary topic books.

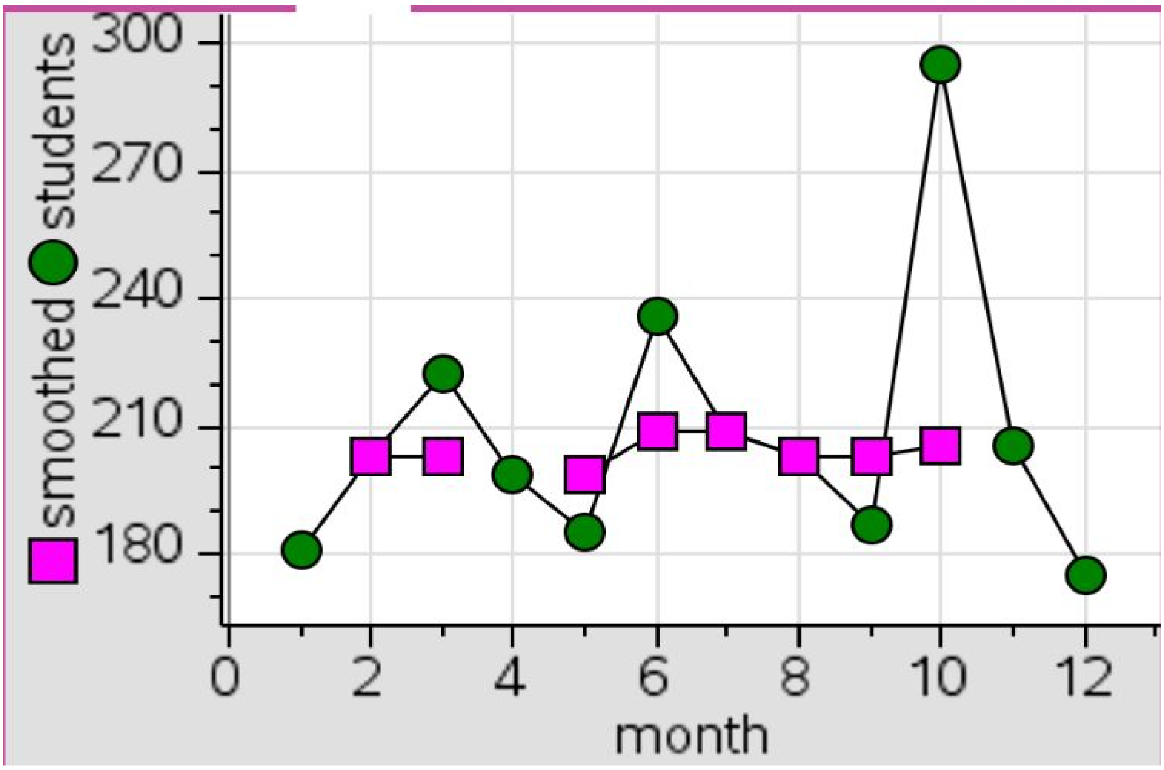
One of the supervisors believes there is an increasing trend in the number of students accessing the one-on-one help sessions and has been gathering data in 2019.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sep	Oct	Nov	Dec
Number of Students attending one-on-one sessions.	181	203	222	199	185	236	209	203	187	295	205	175

The time series plot for the 2019 data is provided below:

Raw data - the number of students time series

The three median smoothed number of students time series



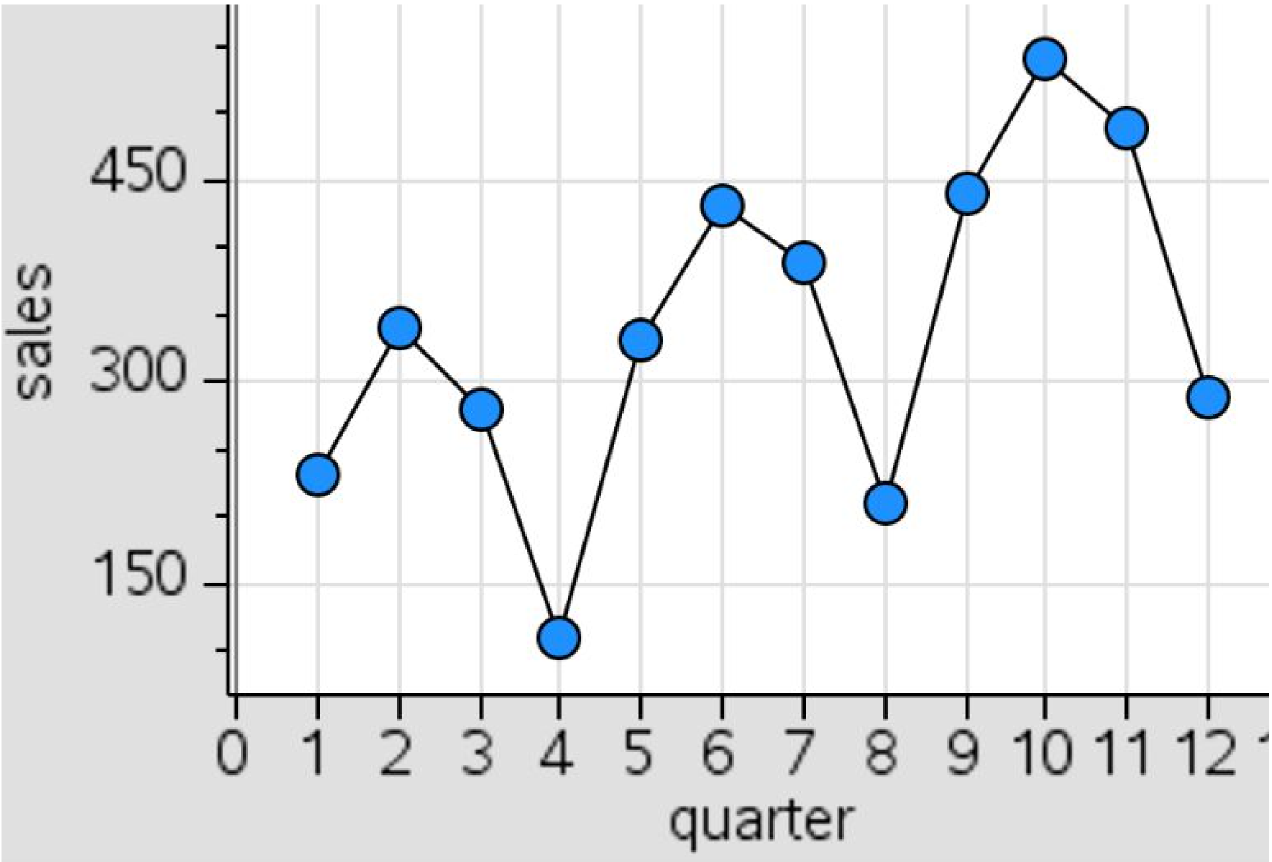
15. Complete the three median smoothed time series plot on the above graph. (2 marks)

16. Does the three median smoothed time series support the supervisor's belief that there has been an increasing trend in the number of students accessing the one-on-one help sessions during the previous year? Briefly explain your answer. (1 mark)

The Maths Enhancement Centre has also been monitoring the sales of their topic summary books. The following table shows the number of sales of the topic summary books in each quarter over the last 3 years.

	Quarter 1	**Quarter 2**	**Quarter 3**	**Quarter 4**	**Quarterly Average**
2017	230	340	280	110	240
2018	330	430	390	210	
2019	440	540	490	290	
Seasonal Index	0.98		1.14	0.58	

The time series plot for this three year period is provided below.



17. Does this time series graph indicate any seasonality in the data? Explain. (1 mark)

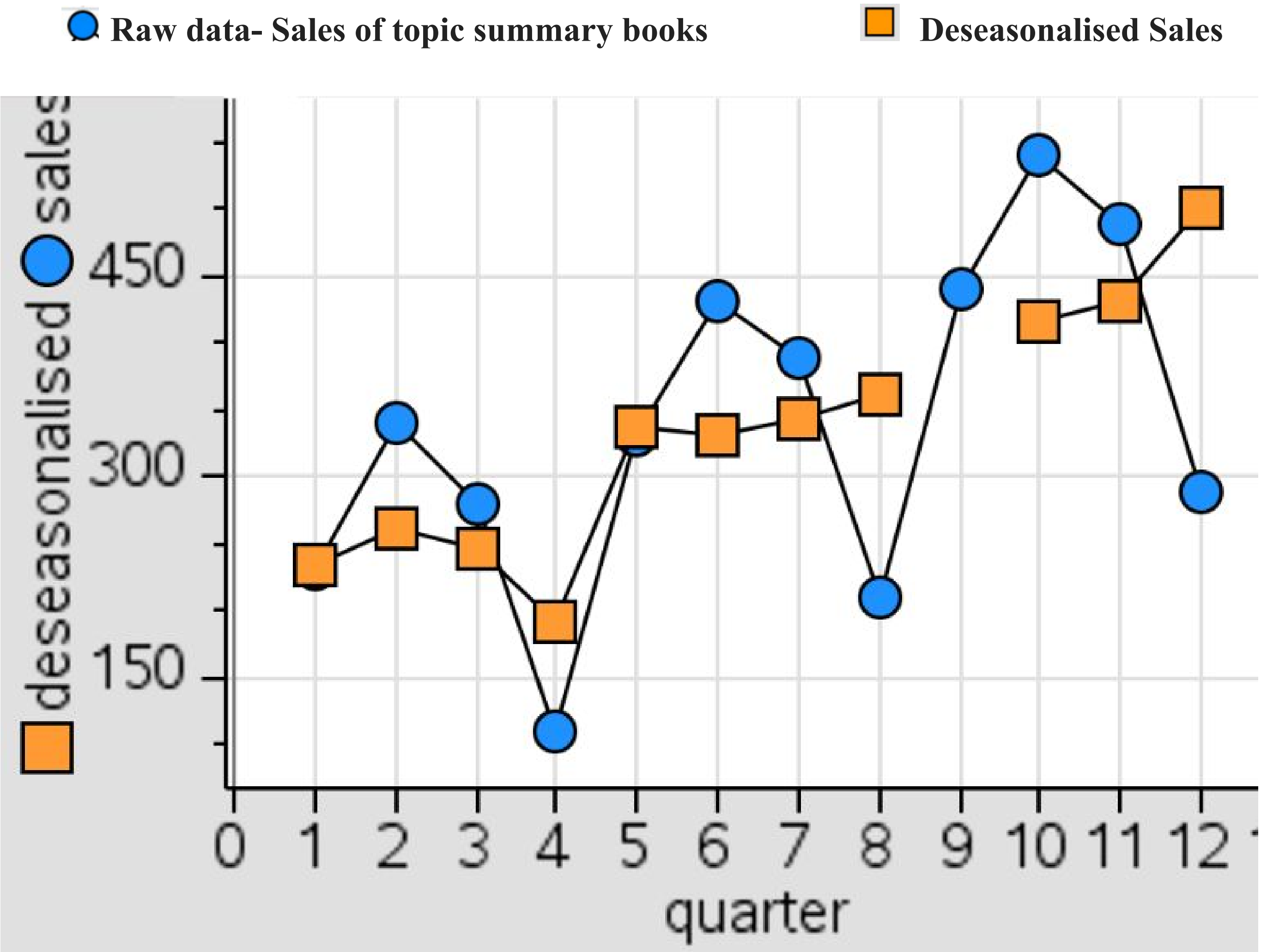
18. Calculate the quarterly averages for 2018 and 2019. Put your answers in the table on the previous page. (1 mark)

19. Calculate the seasonal index for Quarter 2 correct to 2 decimal places. Interpret what this seasonal index means. (2 marks)

20. The following table shows the deseasonalised quarterly topic summary book sales. Find the missing value for Quarter 1, rounding to the nearest book. (1 mark)

	Quarter 1	Quarter 2	Quarter 3	Quarter 4
2017	235	262	246	190
2018	337	331	342	362
2019		415	430	500

21. Using your answer from question 20, complete the following time series of the 2017-2019 deseasonalised quarterly topic summary book sales. **(1 mark)**



22. Comment on the trend displayed by the deseasonalised data. **(1 mark)**

23. Determine the least squares regression equation for the deseasonalised quarterly topic summary book sales. Give your answers to two decimal places. **(2 marks)**

Deseasonalised topic summary book sales = ____________ + __________ × quarter

24. Graph the deseasonalised equation from question 23 on your graph from question 21. Show the calculations for the two coordinate points used to construct the line and mark them in on the graph. (Give your answer to the nearest whole number) (2 marks)

25. Calculate the expected number of topic summary book sales for Quarter 3 in 2020. Show your working out. (Give your answer to the nearest whole number) (2 marks)

END OF TEST - 41 Marks